Staying Power: The Individual Effects of the H-1B Visa Lottery

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Motivation

- Large literature: effects of immigration on U.S. labor market [Card 1990; Borjas 2003; Ottaviano and Peri 2012]

- Common setting: **H-1B visa program**
  - Tied to employer-worker pair; allocated *via lottery*
  - Recent concerns about fraud + rising demand → new $100k fee

- Existing literature: GE, local labor market, firm effects [Bound et al. 2017; Doran et al. 2022; Mahajan et al. 2024]
  - How are firms using the lottery in practice?
  - What happens to workers who lose the lottery?

- **This Project:** New application-level data + linkage strategy → first to estimate **worker-level effects** of winning H-1B lottery

This Paper

Research Question: What are the effects of winning the H-1B lottery on the individual worker?

1. Novel linked panel data: focus on **international students** applying for H-1B
   - Large subset of modern H-1B applicants (40–50%) are recent U.S. graduates
   - Linked panel of 167k H-1B lottery applications to LinkedIn work + location histories
2. Estimate effects of lottery win on indiv. outcomes
3. Investigating robustness and heterogeneity: why are effects small?

This Paper

Research Question: What are the effects of winning the H-1B lottery on the individual worker?

1. Novel linked panel data: focus on **international students** applying for H-1B
2. Estimate effects of lottery win on indiv. outcomes
   - **Preview:** Winning $\uparrow$ U.S. retention by 1.52 pp (1.7%) 2 yrs post-lottery
   - fill in with more results
3. Investigating robustness and heterogeneity: why are effects small?

This Paper

Research Question: What are the effects of winning the H-1B lottery on the individual worker?

1. Novel linked panel data: focus on **international students** applying for H-1B
2. Estimate effects of lottery win on indiv. outcomes
3. Investigating robustness and heterogeneity: why are effects small?
    1. **Measurement Error:** Estimated effects 3-4x larger in higher-confidence linked samples, but still small
    2. **Student visa:** Effects largest for workers near student work authorization expiration, but still small

This Paper

Research Question: What are the effects of winning the H-1B lottery on the individual worker?

1. Novel linked panel data: focus on **international students** applying for H-1B
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Empirical Puzzle: How are lottery losers managing to stay?

Literature Review

- **Effects of H-1B Visa**
  - Aggregate and firm-level impacts on native employment and wages [Kerr et al. 2015; Peri et al. 2015a; Doran et al. 2022; Mahajan et al. 2024], innovation and productivity [Kerr and Lincoln 2010; Peri et al. 2015b; Dimmock et al. 2022; Mayda et al. 2023], and employer monopsony power [Kim and Pei 2022]
  - Global effects on offshoring [Glennon 2024], origin countries [Khanna and Morales 2023], and third-party countries [Brinatti and Guo 2024]
  - **This paper:** first individual-level evidence
- **International Students and U.S. Labor Markets**
  -
  - Offshoring, Indian IT industry, immigration to third countries [Glennon 2024; Khanna and Morales 2023; Brinatti and Guo 2024]
  - Brain drain and return migration [Amanzadeh et al. 2024; Adda et al. 2022]
  - **This paper:** quantifies effect of H-1B cap on individual workers globally
- **Probabilistic Record Linkage** [Fellegi and Sunter 1969; Bailey et al. 2020; Enamorado

Outline

1. Introduction
2. Background and Context
3. Data and Linkage
4. Results
5. Conclusion

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What is an H-1B Visa?

- Temporary work visa (6 years) for workers in 'specialty occupations'
  - Requires highly specialized knowledge + at least a bachelor's degree
- Employer-sponsored: firm must apply on worker's behalf
  - Portable between employers, but no self-sponsorship
  - Employer can sponsor worker for green card (but application tied to employer)
- Capped at 85,000 visas per year since 2004
  - 65,000 regular cap + 20,000 advanced degree exempt (ADE) for U.S. M.A.+
  - Demand has exceeded cap every year since 2013
- Since 2013: visas allocated via **lottery** each spring
  - Results in April, visa valid starting October

The H-1B Lottery: Key Facts

- FY2021: USCIS moved to electronic registration ($10 fee vs. $500+ full petition)
  - **Critical for this paper:** USCIS began keeping records of lottery *losers*
  - Prior data: only winners observed
- Electronic system enabled fraud: duplicate applications via different employers
  - Outsourcing/staffing firms 'overapply' — treat workers as interchangeable
  - I exclude firm-years with any duplicate applicants
- Win rates: ∼35–40% for regular lottery
  - ADE applicants entered into separate lottery first (higher win rate)
- **Identification:** within firm-year, lottery selection is as good as random

Optional Practical Training: The Key Bridge

- F-1 (student) visa holders may work after graduation under OPT
  - Standard OPT: 12 months of authorized employment
  - STEM extension: up to 36 months total for STEM graduates
- OPT has become dominant pathway into H-1B
  - 50–70% of recent H-1B approvals go to workers already in U.S.
  - 94% of F-1→H-1B transitions use OPT
- **Common trajectory:** graduate → OPT → apply for H-1B each spring (up to 3 tries)
  - Worker already employed at sponsoring firm during OPT
  - Firm retains worker if they win; worker faces uncertainty if they lose
- **Key insight:** pre-lottery employment relationship enables linkage

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Data Sources

- **H-1B Application Data** (FOIA via Bloomberg, FY2020–2023)
  - All applicants: country of birth, birth year, sex, employer name/location
  - Winners only: current visa type, education level/field, job title/salary
  - 1.8 million applications; first dataset with both winners and losers
- **Revelio Labs LinkedIn Data**
  - Profiles (894m individuals): full name, education history, work history
  - Positions (1.7b records): employer, title, start/end dates, location, imputed salary
  - Cross-sectional snapshot; not directly identified with H-1B status
- **Challenge:** H-1B data is deidentified — no names, no unique identifiers
  - Standard text matching infeasible

Linkage: Using Pre-Lottery Employment as an Anchor

Key insight: restrict candidates to workers already at sponsoring firm before lottery

1. **Employer crosswalk:** fuzzy match H-1B firms to Revelio firms (93% match rate)
   - Exclude firm-years with 10+ applications or any duplicate applicants
2. **Revelio imputation:** estimate country of birth, birth year, sex, graduation year
   - Country: from education history location (high school) + name-based model
   - Birth year: inferred from education timing
   - Sex: name-based probabilistic imputation
3. **Candidate scoring:** for each applicant, score LinkedIn workers at same firm
   - Weights: country of birth (70%), birth year (20%), sex (10%)
   - Convert scores to within-applicant weights summing to 1

Result: weighted panel of 167,598 linked applications

Sample Variants

- **One-to-one** (preferred): optimal bipartite matching (N = 64,781)
  - Each applicant assigned to at most one Revelio candidate
- **Baseline:** 167,598 linked applications
  - Restrict to U.S.-educated applicants (not U.S. high school); 70,580 with firm-year FE
- **Rare Country:** exclude high-multiplicity countries (N = 10,440)
  - Drop applicants from Southern Asia, East Asia, UK, Canada, Australia
  - Lower candidate pool multiplicity $\rightarrow$ sharper individual identification
- **Strict (Q25 / Q50 / Q75):** score threshold samples
  - Restrict to applicants whose best match score exceeds the 25th, 50th, or 75th percentile
  - Higher confidence matches; trade off sample size for precision

Sample Variants: Match Validation

Higher-confidence samples show better agreement between FOIA fields and matched Revelio profiles.

Match Quality Table

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Empirical Specification

For applicant i at firm c in lottery year t, estimate:

$$y_{ict}^{s} = \beta_0 + \beta_1 Win_{ict} + \beta_2 ADE_{it} + \gamma_{ct} + \varepsilon_{ict}^{s}$$

- Win_{ict}: lottery selection indicator
- ADE_{it}: advanced degree exempt (separate, higher-odds lottery)
- γ_{ct}: firm-by-lottery-year fixed effects
- Outcomes measured s years post-lottery; weights from probabilistic linkage
- Standard errors clustered by firm

Identification: within firm-year, lottery selection is quasi-random conditional on ADE status

Winning the Lottery Stabilizes U.S. Presence

	In U.S.	In home ctry	Non-US non-home	Same firm	New pos. (cond.)	New educ.
	(1)	(2)	(3)	(4)	(5)	(6)
winner	0.0116^{***}	-0.0060^{***}	-0.0033^{***}			
	(0.0027)	(0.0015)	(0.0011)			
Control mean	0.950	0.019	0.010			
N	64,781	64,781	64,781			

OLS estimates; one-to-one linked sample ($N = 64{,}781$); firm-by-lottery-year FE; SEs clustered by firm. $t = 1$ year post-lottery. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Additional outcomes

Winning the Lottery Stabilizes U.S. Presence and Firm Retention

	In U.S.	In home ctry	Non-US non-home	Same firm	New pos. (cond.)	New educ.
	(1)	(2)	(3)	(4)	(5)	(6)
winner	0.0116^{***}	-0.0060^{***}	-0.0033^{***}	0.0264^{***}	-0.0146^{*}	
	(0.0027)	(0.0015)	(0.0011)	(0.0053)	(0.0075)	
Control mean	0.950	0.019	0.010	0.743	0.448	
N	64,781	64,781	64,781	64,781	45,209	

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Winning the Lottery Stabilizes U.S. Presence, Firm Retention, and Education

	In U.S.	In home ctry	Non-US non-home	Same firm	New pos. (cond.)	New educ.
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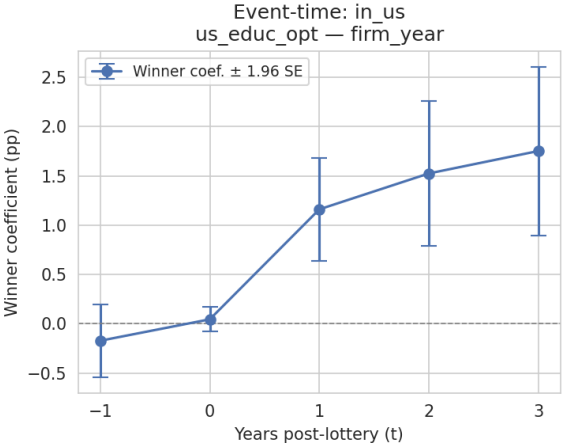
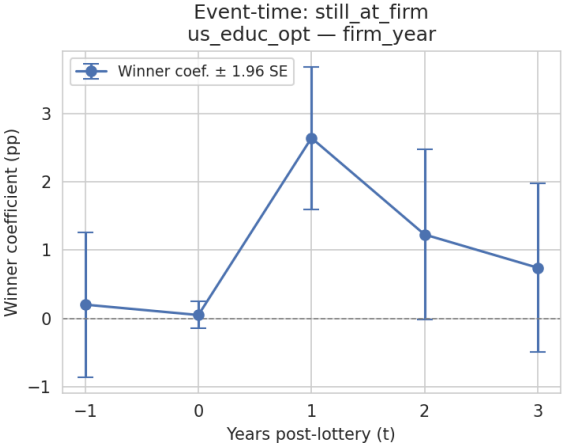
Additional outcomes

Effect on U.S. Presence Grows as OPT Runway Shrinks



Raw means

Winners Remain Locked Into Sponsoring Firm Over Time



Raw means

Taking Stock: Why Are Effects Small?

- Effects are statistically significant but moderate in the one-to-one sample (1.52 pp U.S. presence)
- **Reason 1:** Measurement error attenuates estimates
  - Probabilistic linkage mixes correct and incorrect matches
  - **How do effects vary by linkage quality?**
- **Reason 2: OPT workers have fallback options**

Effects Robust Across Higher-Confidence Linked Samples

	In U.S.	Same firm	New pos. (cond.)	New educ.
	(1)	(2)	(3)	(4)
One-to-one ($N = 64{,}781$)	0.0116^{***} (0.0027)	0.0264^{***} (0.0053)	-0.0146^{*} (0.0075)	-0.0042 (0.0059)

$\hat{\beta}_1$ *(winner effect) with SEs in parentheses; firm-by-lottery-year FE; SEs clustered by firm.* $t = 1$ *year post-lottery.*
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Control means

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Baseline ($N = 70{,}580$)	0.0023^{***} (0.0008)	0.0058^{***} (0.0017)	-0.0010 (0.0020)	0.0040^{**} (0.0020)

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Baseline ($N = 70{,}580$)	0.0023^{***} (0.0008)	0.0058^{***} (0.0017)	-0.0010 (0.0020)	0.0040^{**} (0.0020)
Strict Q25 ($N = 40{,}165$)	0.0103^{***} (0.0026)	0.0253^{***} (0.0048)	-0.0062 (0.0063)	-0.0073 (0.0059)
Strict Q50 ($N = 27{,}765$)	0.0146^{***} (0.0034)	0.0328^{***} (0.0060)	-0.0169^{**} (0.0081)	-0.0085 (0.0075)
Strict Q75 ($N = 13{,}894$)	0.0142^{***} (0.0049)	0.0408^{***} (0.0086)	-0.0336^{***} (0.0119)	0.0041 (0.0113)

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Rare Country ($N = 10{,}440$)	0.0083^{**} (0.0033)	0.0221^{***} (0.0061)	-0.0097 (0.0072)	-0.0087 (0.0069)

$\hat{\beta}_1$ (winner effect) with SEs in parentheses; firm-by-lottery-year FE; SEs clustered by firm. $t = 1$ year post-lottery.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Control means

Taking Stock: Why Are Effects Small?

- Effects are statistically significant but moderate in the one-to-one sample (1.16 pp U.S. presence)
- **Reason 1: Measurement error attenuates estimates**
- **Reason 2: OPT workers have fallback options**
  - Selecting on OPT $\implies$ applicants can continue to work through OPT!
  - **Prediction:** effects largest when applicants have no OPT remaining
    - Workers 1 yr post-graduation: nearing end of initial 12-month OPT
    - Workers 3 yrs post-graduation: nearing end of STEM-OPT extension
  - Effects larger for firms less likely to re-sponsor workers (*no-rep*) vs. *high-rep* firms
  - **How do effects vary by years since graduation and employer type?**

Effects Strongest for Workers Near OPT Expiration

	0 yrs post-grad	1 yr post-grad	2 yrs post-grad	3 yrs post-grad	No rep.	High rep.
	(1)	(2)	(3)	(4)	(5)	(6)
winner	0.0210	0.0324**	0.0395**	0.0446**	0.0221	0.0029
	(0.0195)	(0.0138)	(0.0164)	(0.0194)	(0.0211)	(0.0064)
Control mean	0.893	0.880	0.870	0.875	0.889	0.920
N	1,980	6,029	4,728	3,043	2,605	19,322

Outcome: In U.S. (2 years post-lottery). Cols (1)–(4): subgroups by years since graduation at time of lottery application. Cols (5)–(6): by employer repetition. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Summary of Results

- **Main effects:** winning H-1B raises U.S. presence (+1.52 pp) and firm retention (+1.23 pp)
  - Winners less likely to leave for a different employer (−1.23 pp)
  - Winners less likely to appear in home country (−0.74 pp)
- **Robustness:** effects consistent across higher-confidence samples
  - U.S. presence reaches 2.0 pp (Q75); firm retention 1.45 pp
  - Baseline effects smaller — consistent with attenuation bias from probabilistic linkage
- **Heterogeneity:** effects driven by workers near OPT expiration
  - Patterns at 2 and 4 yrs post-graduation consistent with OPT expiration dates
- **Interpretation:** H-1B as *stability lottery* — reduces disruption, locks workers into firm

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Conclusion

- **Contribution 1:** first individual-level causal estimates of H-1B lottery effects
  - Winning stabilizes U.S. presence and firm retention; reduces employer mobility
  - For OPT workers, H-1B is a stability mechanism, not an entry gate
- **Contribution 2:** novel probabilistic linkage for deidentified immigration records
  - Pre-lottery employment as anchor reduces candidate space from millions to hundreds
- **Policy implication:** modest average effects $\neq$ unimportant policy
  - Repeated reapplication costs, employer dependence, and uncertainty have welfare costs
  - Reforms targeting portability and OPT-to-H1B transitions could reduce disruption
- **Next steps:**
  - Wage and compensation effects
  - Productivity outcomes via GitHub data
  - Return migration and third-country reallocation

Additional Outcomes: One-to-One Sample

	Profile in U.S.	Cont. education	Activity update diff	Frac. null end date
	(1)	(2)	(3)	(4)
winner	—	0.0040	0.7620^{***}	-0.0129^{**}
	—	(0.0059)	(0.2922)	(0.0056)
Control mean	—	0.663	17.011	0.408
N	—	64,781	64,781	64,781

OLS estimates; one-to-one linked sample (N = 64,781); firm-by-lottery-year FE; SEs clustered by firm. $t = 1$ year post-lottery. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Return

Robustness: Control Means by Sample

	In U.S.	Same firm	New pos. (cond.)	New educ.
One-to-one	0.950	0.743	0.448	0.340
Baseline	0.956	0.743	0.388	0.301
Strict Q25	0.938	0.741	0.442	0.452
Strict Q50	0.934	0.740	0.449	0.466
Strict Q75	0.937	0.746	0.455	0.473
Rare Country	0.948	0.749	0.413	0.304

Control group means; $t = 1$ year post-lottery.

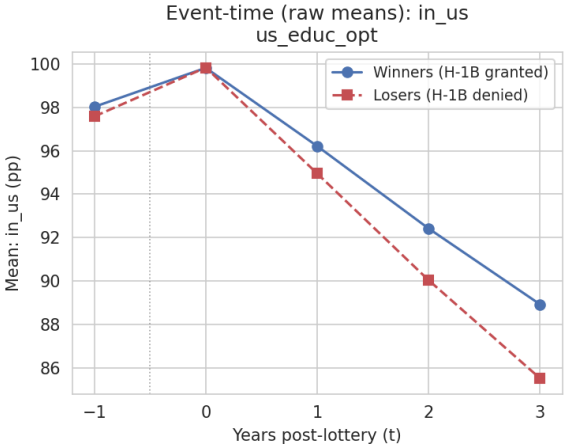
Return

Higher-Quality Samples Have Better Match Validation

Dimension	FOIA cov.	U.S.-educ.	Pre-filtered	One-to-one	Strict Q50
Education level	26–33%	62.0%	61.7%	73.9%	85.7%
Institution (ADE)	—	34.4%	45.9%	46.6%	49.2%
Field exact	100%	19.6%	20.0%	27.9%	33.7%
Job title	100%	4.0%	3.7%	6.1%	9.0%
Ed. + any other	—	34.9%	34.6%	47.1%	60.3%
N (applications)		106,819	25,231	101,797	47,881

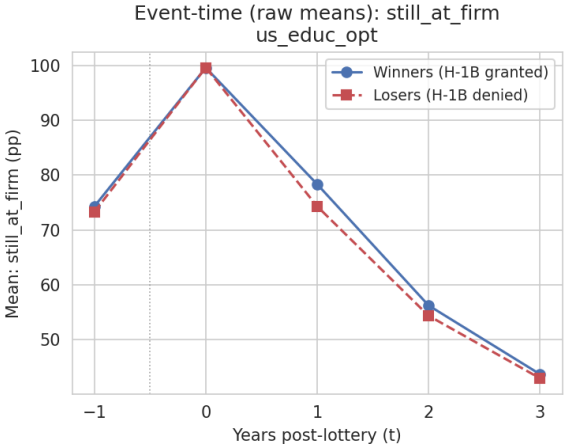
Agreement rates between FOIA petition fields and matched Revelio profiles, by sample.

Raw Means: U.S. Presence by Lottery Outcome



Return

Raw Means: Firm Retention by Lottery Outcome



Return